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Para Inf

$$* \quad X_1, \dots, X_n \stackrel{iid}{\sim} f(x|\theta)$$

$\delta(\underline{X})$ UMVUE of $\psi(\theta)$. Then $\delta(\underline{X})$ is symmetric in $\underline{X}$ i.e.

$$\delta(X_1, \dots, X_n) = \delta(X_{i_1}, \dots, X_{i_n}) \quad \forall \text{ permutation } \{i_1, \dots, i_n\}$$

$$\delta^* = \frac{1}{n!} \sum_{(i_1, \dots, i_n) \in \mathcal{S}} \delta(X_{i_1}, \dots, X_{i_n}) \longrightarrow \text{symmetric}$$

— If T is complete sufficient statistic, then T must be minimal sufficient for θ (R.R. Bahadur)

Pf. Let U = any suff stat for θ . To show, T is some fn. of U

For generality let $T = (T_1, \dots, T_k)$ be a k -dim statistic

Let's define $S_i(T) = (1 + e^{-T_i})^{-1}$ for $i=1, \dots, k$
(can consider any 1-1 bounded fn.)

$$\text{Define } X_i(U) = E[S_i(T)|U]$$

$$Y_i(T) = E[X_i(U)|T]$$

$$\text{If } S_i(T) = X_i(U) \Rightarrow T_i = S_i^{-1}(X_i(U))$$

$$1. \quad S_i(T) = Y_i(T)$$

$$\text{b/c } E[Y_i(T)] = E[E(X_i(U)|T)] = E X_i(U)$$

$$\stackrel{||}{=} E[E[S_i(T)|U]] = E S_i(T)$$

$$E_\theta [S_i(T) - Y_i(T)] = 0 \quad \forall \theta \Rightarrow S_i(T) = Y_i(T) \quad \text{w.p. 1}$$

— * (completeness)

2. To show $Y_i(T) = X_i(U)$

$$\begin{aligned} V(Y_i(T)) &= E[V(Y_i(T)|U)] + V[E(Y_i(T)|U)] \\ &= E[V(Y_i(T)|U)] + V(X_i(U)) \quad \text{by } * \\ &= E[V(Y_i(T)|U)] + E[V(X_i(U)|T)] + V[\underbrace{E(X_i(U)|T)}_{Y_i(T)}] \end{aligned}$$

$$\Rightarrow E[V(Y_i(T)|U)] + E[V(X_i(U)|T)] = 0$$

$$\Rightarrow V(Y_i(T)|U) = 0$$

$$\begin{aligned} Y_i(T) &= E(Y_i(T)|U) \\ &= E(S_i(T)|U) \\ &= X_i(U) \end{aligned}$$

$$\therefore S_i(T) = Y_i(T) = X_i(U)$$

$$\Rightarrow T_i = S_i^{-1}(X_i(U)) \quad \forall i=1, \dots, k$$

$$\Rightarrow T \text{ is a f.n. of } U \quad \square$$

Lower Bounds for Variance of an estimator

1. Cramer Rao Lower Bound (CRLB)

2. Chapman-Robbins Bound

CRLB (Fréchet, Cramer, Rao)

Let $\underline{X} = (X_1, \dots, X_n)$ have pmf/pdf $f(\underline{x}|\theta) : \theta \in \Theta \subseteq \mathbb{R}$
 $\hookrightarrow \Theta$ is open

Suppose following regularity conditions hold

(1) $\mathcal{X}_\theta = \{x: f(x|\theta) > 0\}$ is free of θ

(2) $|\frac{\partial}{\partial \theta} \ln f(x|\theta)| < \infty \forall x \in \mathcal{X} \text{ and } \theta \in \Theta$

(3) For any statistic $h(\underline{x})$ with $E_\theta |h(\underline{x})| < \infty \forall \theta$, we can change integration (w.r.t x) and derivative (w.r.t θ)

$$\frac{\partial}{\partial \theta} \int h(x) f(x|\theta) dx = \int h(x) \frac{\partial}{\partial \theta} f(x|\theta) dx$$

If $\delta(\underline{x})$ is an u.e. of $\psi(\theta)$ with $E_\theta (\delta^2(\underline{x})) < \infty$, $\frac{\partial}{\partial \theta} \psi(\theta)$ exists $:= \psi'(\theta)$ and the Fisher info $I_X(\theta) \in (0, \infty)$ then

$$V(\delta(\underline{x})) \geq \frac{(\psi'(\theta))^2}{I_X(\theta)}$$

pf: $\psi'(\theta) = \frac{\partial}{\partial \theta} E_\theta (\delta(\underline{x})) = \frac{\partial}{\partial \theta} \int \delta(x) f(x|\theta) dx$

$$\stackrel{(3)}{=} \int \delta(x) \frac{\partial}{\partial \theta} f(x|\theta) dx$$

$$= \int \delta(x) \frac{\partial}{\partial \theta} (\ln f(x|\theta)) f(x|\theta) dx$$

$$= E_\theta [\delta(X) \frac{\partial}{\partial \theta} \ln f(X|\theta)]$$

$$= \text{Cov}_\theta(\delta(X), \frac{\partial}{\partial \theta} \ln f(X|\theta)) \quad \text{b/c } E\left[\frac{\partial}{\partial \theta} \ln f(X|\theta)\right] = 0$$

$$(\psi'(\theta))^2 = \text{Cov}^2(\delta(X), \frac{\partial}{\partial \theta} \ln f(X|\theta))$$

$$\leq V(\delta(X)) \underbrace{V\left(\frac{\partial}{\partial \theta} \ln f(X|\theta)\right)}_{= I_X(\theta)}$$

$$\Rightarrow V(\delta(X)) \geq \frac{(\psi'(\theta))^2}{I_X(\theta)} = \text{CRLB} \quad \square$$

Note: (a) If $\text{var } \delta$ attains CRLB, then δ is UMVUE of $\psi(\theta)$

(b) If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f(x|\theta)$ then

$$I_X(\theta) = n I_{X_1}(\theta) \quad \text{then} \quad \text{CRLB} = \frac{(\psi'(\theta))^2}{n I_{X_1}(\theta)} \downarrow n \uparrow$$

Ex. $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bern}(\theta)$

UMVUE of θ is $\bar{X}$

$$V(\bar{X}) = \frac{\theta(1-\theta)}{n}$$

$$\psi(\theta) = \theta$$

$$\therefore (\psi'(\theta))^2 = 1$$

$$I_{X_1}(\theta) = -E\left(\frac{\partial^2}{\partial \theta^2} \ln \theta^X (1-\theta)^{1-X}\right)$$

$$= -E\left(\frac{\partial}{\partial \theta} \left(\frac{X}{\theta} - \frac{1-X}{1-\theta}\right)\right)$$

$$= -E\left(\frac{-X}{\theta^2} - \frac{1-X}{(1-\theta)^2}\right)$$

$$= \frac{\theta}{\theta^2} + \frac{1-\theta}{(1-\theta)^2} = \frac{1}{\theta} + \frac{1}{1-\theta} = \frac{1}{\theta(1-\theta)}$$

$$\text{CRLB} = \frac{1}{n I_{X_1}(\theta)}$$

$$= \frac{\theta(1-\theta)}{n}$$

attained

$\therefore \bar{X}$ is UMVUE

Ex. $X_1, \dots, X_n \sim \text{Pois}(\lambda)$

$$V(\bar{X}) = \frac{\lambda}{n} = \text{CRLB}$$

Ex Let $Y \sim \text{Pois}(\lambda)$ Estimate $\psi(\lambda) = e^{-\lambda}$. Find the UMVUE of $\psi(\lambda)$ and check whether CRLB attains or not

Suppose $\delta(x)$ is an unbiased estimator of $e^{-\lambda}$

$$\Rightarrow \sum_{x=0}^{\infty} \delta(x) \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda}$$

$$\Rightarrow \delta(0) + \delta(1)\lambda + \frac{\delta(2)\lambda^2}{2!} + \dots = 1$$

$$\Rightarrow \delta(0) = 1, \delta(1) = \delta(2) = \dots = 0$$

$\therefore \delta(x) = \mathbb{1}_{\{x=0\}}$ is the only u.e. of $e^{-\lambda}$ ($\overset{\text{UMVUE}}{\text{of } e^{-\lambda}}$)
 $\sim \text{Bern}(e^{-\lambda})$

$$V(\delta(x)) = e^{-\lambda}(1 - e^{-\lambda})$$

$$\begin{aligned} I_X(\lambda) &= -\mathbb{E} \left[\frac{\partial^2}{\partial \lambda^2} \ln \frac{e^{-\lambda} \lambda^x}{x!} \right] = -\mathbb{E} \left[\frac{\partial}{\partial \lambda} \left(-1 + \frac{x}{\lambda} \right) \right] \\ &= \mathbb{E} \left[\frac{x}{\lambda^2} \right] = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda} \end{aligned}$$

$$\text{CRLB} = \frac{e^{-2\lambda}}{1/\lambda} = \lambda e^{-2\lambda}$$

$$e^{-\lambda}(1-e^{-\lambda}) > \lambda e^{-2\lambda}$$

$$\text{b/c } e^{\lambda}-1 > \lambda \quad (\text{true})$$

$$\sum_0 \text{CRLB} < V(\underbrace{\delta(X)}_{\text{UMVUE}}) \quad (\text{Not attained})$$

$$X \sim U(0, \theta) \quad f(x|\theta) = \begin{cases} 1/\theta & \text{if } 0 < x < \theta \\ 0 & \text{o/w} \end{cases}$$

$2X$ is an u.e. of θ

$$\frac{\partial}{\partial \theta} \ln f(x|\theta) = \frac{\partial}{\partial \theta} (-\ln \theta) = -\frac{1}{\theta}$$

$$\Rightarrow E\left(\left(\frac{\partial}{\partial \theta} \ln f(x|\theta)\right)^2\right) = \frac{1}{\theta^2}$$

$$\text{CRLB} = \theta^2$$

$$\text{But } V(2X) = 4 \frac{\theta^2}{12} = \frac{\theta^2}{3}$$

When does "=" in CRLB?

Let $\delta(X)$ be u.e. of $\psi(\theta)$ with $E\delta(X)^2 < \infty$

$$\text{we had } \text{Cov}^2(\delta(X), \frac{\partial}{\partial \theta} \ln f(x|\theta)) \leq V(\delta(X)) V(\frac{\partial}{\partial \theta} \ln f(x|\theta))$$

$$\Rightarrow "=" \text{ holds if } \delta(X) = a(\theta) \frac{\partial}{\partial \theta} \ln f(x|\theta) + b(\theta)$$

$$E\delta(X) = \psi(\theta) = a(\theta) \underbrace{E\left(\frac{\partial}{\partial \theta} \ln f(x|\theta)\right)}_0 + b(\theta) = b(\theta)$$

$$\therefore \delta(X) - \psi(\theta) = a(\theta) \frac{\partial}{\partial \theta} \ln f(x|\theta) \quad \text{Let } K(\theta) = \frac{1}{a(\theta)}$$

$$\Rightarrow K(\theta)(\delta(X) - \psi(\theta)) = \frac{\partial}{\partial \theta} \ln f(x|\theta)$$

$$\Rightarrow \ln f(x|\theta) = \int K(\theta) \delta(X) d\theta - \int K(\theta) \psi(\theta) d\theta + h(x)$$

$$f(x|\theta) = h(x) e^{\delta(X) \underbrace{\int K(\theta) d\theta}_{\eta(\theta)} - \underbrace{\int K(\theta) \psi(\theta) d\theta}_{B(\theta)}}$$

Note: (1) If an u.e. $\delta(X)$ of $\psi(\theta)$ has to attain CRLB, $\delta(X)$ must be sufficient stat for θ .

(2) Suppose $\delta(X)$ is a biased estimator of $\psi(\theta)$

$$B_{\theta}(\delta) = \mathbb{E}(\delta(X) - \psi(\theta))$$

$$\mathbb{E}\delta(X) = B_{\theta}(\delta) + \psi(\theta) = \psi^*(\theta)$$

$$V(\delta(X)) \geq \frac{\left(\frac{\partial}{\partial \theta} \psi^*(\theta)\right)^2}{I_X(\theta)} \quad \text{MSE}_{\theta}(\delta) = V_{\theta}(\delta) + (B_{\theta}(\delta))^2$$

$$\text{MSE}_{\theta}(\delta) - (B_{\theta}(\delta))^2$$

$$\Rightarrow \text{MSE}_{\theta}(\delta) \geq \frac{\left(\frac{\partial}{\partial \theta} B_{\theta}(\delta) + \psi'(\theta)\right)^2}{I_X(\theta)} + (B_{\theta}(\delta))^2$$

Defⁿ: (1) Let $X \sim f(x|\theta)$, $\theta \in \Theta \subseteq \mathbb{R}$ satisfy all conditions of CR inequality. An u.e. $\delta(X)$ is called the "most efficient" if $\delta(X)$ attains CRLB

(2) If all conditions of CR Inequality hold, then efficiency of an u.e. $\delta(X)$ of $\psi(\theta)$ is defined as

$$e_{\theta}(\delta) := \frac{\text{CRLB}}{V_{\theta}(\delta)} = \frac{(\psi'(\theta))^2}{I_X(\theta) V_{\theta}(\delta)} \leq 1$$

(3) If all conditions of CR Ineq. hold, then u.e. $S(X)$ of $\psi(\theta)$ is called "asymptotically most efficient" if $\lim_{n \rightarrow \infty} e_{\theta}(S) = 1$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Find the efficiency of

$$S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2 \quad (\text{UMVUE of } \sigma^2)$$

$$V(S^2) = \frac{2\sigma^4}{n-1}$$

$$\psi(\sigma^2) = \sigma^2$$

$$\begin{aligned} \text{CRLB: } I_{X_1}(\sigma^2) &= \frac{1}{2\sigma^4} & \therefore \text{CRLB} &= \frac{\left(\frac{0}{\sigma^2} \sigma^2\right)^2}{I_X(\sigma^2)} \\ & & &= \frac{1}{n/2\sigma^4} = \frac{2\sigma^4}{n} \end{aligned}$$

$$\begin{aligned} \text{efficiency} \quad e_{\sigma^2}(S^2) &= \frac{2\sigma^4}{n} \frac{n-1}{2\sigma^4} \\ &= \frac{n-1}{n} \longrightarrow 1 \end{aligned}$$

Hence S^2 is asymptotically most efficient.